



Student Name:

Student ID:

Answer Key for Question #1

Question #1:

[Marks 45]

- a) Simplify the following logical expression using the laws of logic. You must state the name of the law used in each step: [15 pts]

$$\neg(p \rightarrow q) \vee (p \wedge r)$$

Step	Expression	Law Used
1	$\neg(\neg p \vee q) \vee (p \wedge r)$	Conditional Law
2	$(p \wedge \neg q) \vee (p \wedge r)$	De Morgan's Law
3	$p \wedge (\neg q \vee r)$	Distributive Law

- c) Show that the following expression is a Tautology by proving the logical equivalence using a truth table: [25 pts]

$$((p \rightarrow q) \wedge \neg q) \rightarrow \neg p \equiv T$$

p	q	$p \rightarrow q$	$\neg q$	$(p \rightarrow q) \wedge \neg q$	$\neg p$	$((p \rightarrow q) \wedge \neg q) \rightarrow \neg p$
T	T	T	F	F	F	T
T	F	F	T	F	F	T
F	T	T	F	F	T	T
F	F	T	T	T	T	T

Conclusion: Since the final column contains only T (True) values for all possible truth values of p and q , the expression is a Tautology.

- b) Which Rule of Inference is represented by the tautology in **part (b)**? Write its formal argument form. [5 pts]

Name of the Rule Inference: Modus Tollens

Formal Argument Form:

$$p \rightarrow q$$

$$\neg q$$

$$\therefore \neg p$$



Computer Science Department
COMP233-Discrete Mathematics/ Spring 2026
Section #4 - Quiz #1 (14 April): Chapters 1 & 2

Student Name:

Student ID:

Answer Key for Question #2

Question #2:

[Marks 55]

- a) Consider the following statement regarding the Birzeit University academic system: [35 pts]

Concept	Logical Form	Verbal Statement (Optional)
1. Formalization	$\forall x, (M(x) \vee R(x)) \rightarrow P(x)$	(Original Statement)
2. Converse	$\forall x, P(x) \rightarrow (M(x) \vee R(x))$	For every student, if they enter the correct password on Ritaj, then they can check their marks or register for the course.
3. Inverse	$\forall x, \neg(M(x) \vee R(x)) \rightarrow \neg P(x)$	For every student, if they do not check their marks and do not register for the course, then they did not enter the correct password.
4. Contrapositive	$\forall x, \neg P(x) \rightarrow (\neg M(x) \wedge \neg R(x))$	For every student, if they do not enter the correct password on Ritaj, then they cannot check their marks and cannot register for the course.
5. Negation	$\exists x, (M(x) \vee R(x)) \wedge \neg P(x)$	There is a student who checks their marks or registers for the course, but did not enter the correct password on Ritaj.

- b) Provide the direct logical equivalent or operation for each of the following (Write the formal expression): [20 pts]

- 1) Negation of Nested Quantifiers: $\exists x \forall y, \neg P(x, y)$, 2 Arity or Binary, because the predicate takes 2 variables x and y.
- 2) The Contrapositive of the Converse: $\neg p \rightarrow \neg q$ (Or the inverse)
- 3) Inference Rule (Transitivity):
 $p \rightarrow q$
 $q \rightarrow r$
 $\therefore p \rightarrow r$